INCH-POUND

MIL-DTL-83413/7F
w/AMENDMENT 3
21 December 2018
SUPERSEDING
MIL-DTL-83413/7F
w/AMENDMENT 2
6 October 2016

DETAIL SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: GROUNDING CLAMP CONNECTOR FOR TYPES I AND III GROUNDING ASSEMBLIES, ALUMINUM CLIP, ELECTRICAL

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.

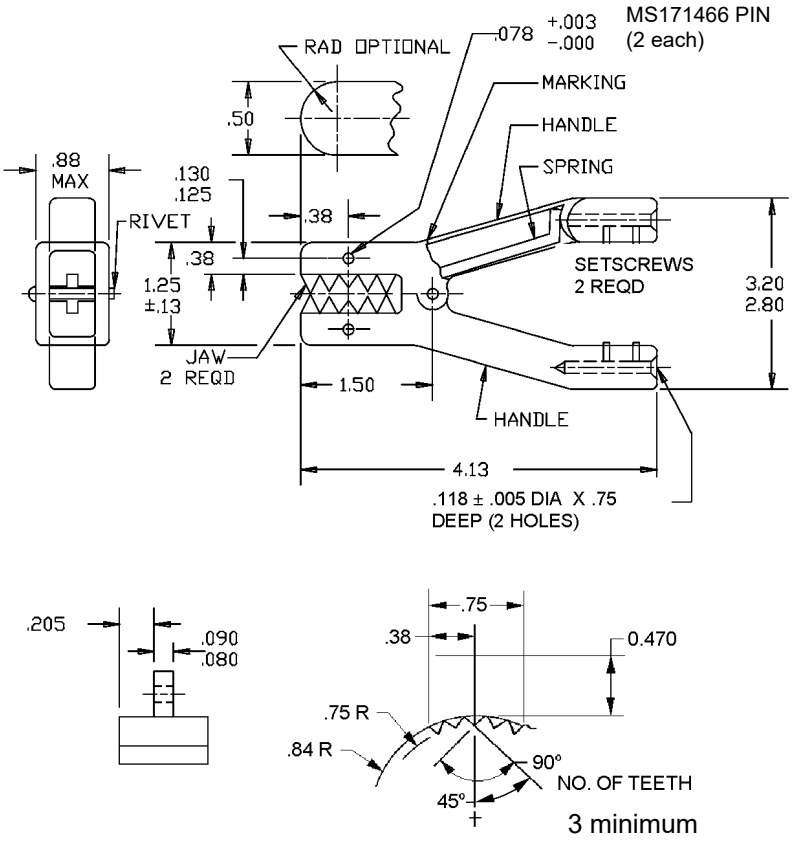


FIGURE 1. Grounding clamp, for types I and III grounding assemblies.

AMSC N/A

FSC 5999

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Inches	mm	Inches	mm	Inches	mm
.003	0.08	.113	2.87	.75	19.1
.004	0.10	.125	3.18	.84	21.3
.010	0.25	.130	3.30	.88	22.4
.031	0.79	.130	3.30	1.25	31.4
.062	1.57	.203	5.16	1.50	38.1
.078	1.98	.205	6.35	2.80	76.1
.080	2.03	.380	9.70	3.20	81.3
.090	2.29	.500	12.70	4.13	104.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Surface toughness: If cast, smooth cast finish 125 or 250 all over, except machine surface 63.
4. Remove all burrs and sharp edges .010 (0.25 mm) to .020 (0.51 mm) radius.
5. Unless otherwise specified, tolerance is ±.03 (0.8 mm) on decimals and ± 1° on angles.

FIGURE 1. Grounding clamp, for types I and III grounding assemblies - Continued.

REQUIREMENTS:

Material:

Handle: The handle material shall be of aluminum or aluminum alloy and shall be in accordance with ASTM B26/B26M, ASTM B85/B85M, ASTM B108/B108M, ASTM B209, ASTM B211, ASTM B221 and ASTM B241/B241M.

Spring: The spring material shall be .125 diameter music wire and meet the spring requirements of MIL-DTL-83413.

Set screw: The set screw shall meet the requirements of AIA/NASM51021, and shall be passivated in accordance with SAE-AMS 2700.

Jaw: The jaw material shall be steel and meet the material requirements of MIL-DTL-83413.

Alternate jaw material: MPIF FX - 1008-110HT in accordance with Metal Powder Industries Federation, 105 College Road East, Princeton, New Jersey 08540.

Jaw teeth inner radius: The .75 R dimension on figure 1 shall be measured from the (point, apex, tip) on the first tooth to the (point, apex, tip) on the last tooth.

Clamp connector resistance: Shall not exceed 500 milliohms when tested in accordance with MIL-DTL-83413.

Part or Identifying Number (PIN): M83413/7-1.

Salt atmosphere (corrosion): Not applicable.

NOTE: This aluminum part will not pass a Salt Spray test. If your application demands salt atmosphere resistance use PIN M83413/12-1, see MIL-DTL-83413/12.

QUALIFICATION: Qualification is not required. First article inspection shall consist of a visual and mechanical inspection in accordance with MIL-DTL-83413 and the requirements stated in this specification sheet. First article inspection shall consist of the inspections specified in table I.

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A QPL manufacturer that periodically performs group A and B inspections on Grounding clamps in accordance with MIL-DTL-83413, is not required to perform group II and group III First Article inspections on Grounding clamps. Group I inspections are still required.

Samples for first article inspection shall be representative of the products proposed to be furnished to this specification. Test samples shall consist of 100% testing, up to 12 grounding clamps. Test samples shall be examined and tested in accordance with MIL-DTL-83413 and this specification. Test results shall be submitted to the acquiring activity.

TABLE I. First article inspection.

Inspection	Plug	Receptacle	Clamp connector	Jumper assembly	Requirement paragraph	Test Paragraph
Group I						
Visual and mechanical Inspection			X		3.1, 3.3, 3.4 3.6, 3.8	4.7.1
Detent action					3.5.1	4.7.2
Handgrip force			X		3.5.2	4.7.3
Side loading					3.5.3	4.7.4
Torque					3.5.4	4.7.5
Engagement					3.5.5	4.7.6
Contact resistance					3.5.6.1	4.7.7.1
Clamp connector resistance			X		3.5.6.2	4.7.7.2
Assembly resistance					3.5.6.3	4.7.7.3
Group II						
Cable attachment			X		3.5.7	4.7.8
Clamp release force (initial)			X		3.5.8.1	4.7.9
Clamp release force (after conditioning)			X		3.5.8.2	4.7.9
Clamping life			X		3.5.9	4.7.10
Vibration					3.5.10	4.7.11
Durability					3.5.11	4.7.12
Flexibility					3.5.13	4.7.14
Tensile strength					3.5.12	4.7.13
Crush					3.5.14	4..7.15
Group III						
Humidity			X		3.5.15	4.7.16
Temperature cycling			X		3.5.16	4.7.17
Drop			X		3.5.17	4.7.18
Altitude – low temperature					3.5.18	4.7.19
Dust					3.5.19	4.7.20
Fluid immersion					3.5.20	4.7.21
Salt spray					3.5.22	4.7.23
Group IV						
Lightning current					3.5.23	4.7.24

1/ - X indicates that a test is applicable.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-83413, this document references the following:

MIL-DTl-83413/12
AIA/NASM51021
MPIF FX-1008-110HT
SAE-AMS 2700
ASTM B26/B26M
ASTM B85/B85M
ASTM B108/B108M
ASTM B209
ASTM B211
ASTM B221
ASTM B241/B241M

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5999-2018-009)

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force - 02

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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